

NISHANTH AYYALASOMAYAJULA

AI Engineer

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SUMMARY

Applied AI engineer with 3+ years taking LLM systems from prototype to production - spanning RAG pipelines, MCP-native agentic architectures, governed AI workflows, and MLOps. Proven at scale: **1,500+ users, 30,000+ monthly interactions, 98.21% NLP accuracy**. Salesforce Certified AI Specialist. M.S. Information Technology, Florida State University (GPA 4.0/4.0).

EXPERIENCE

Lead AI Engineer · [Starteryou](#)

Jul 2025 – Feb 2026 ·
Tallahassee, FL (Remote)

- Led end-to-end delivery of a production AI platform - FastAPI backend, PostgreSQL, Azure OpenAI, vector search, Terraform IaC, and CI/CD - scaling to **1,500+ users and 30,000+ monthly interactions**.
- Automated 5+ manual operations workflows into AI-driven pipelines, reducing average execution time by an estimated **~40%** and cutting key cross-team delivery bottlenecks.
- Integrated AI into fraud and payment-review workflows, reducing manual review volume by **35%** and enabling faster, more consistent decisions.
- Established model evaluation and monitoring workflows for system stability, enabling safer feature rollouts across a live production user base.
- Mentored 3 engineers on AI system design and delivery practices; improved team execution consistency across cross-functional initiatives.

Applied NLP Researcher · [Florida State University](#)

Jan 2025 – May 2025 ·
Tallahassee, FL (On Site)

- Developed a multiclass healthcare misinformation classifier across **118,750 labeled records** (True / False / Mixture / Fabricated); designed for public-health screening and downstream automation.
- Benchmarked 5 model families; achieved **98.21% accuracy and 0.99 F1-score** with Linear SVM - strong enough baseline for production deployment.
- Extended to transformer architectures (DistilBERT, TinyBERT, BERT, Qwen); automated experiment logging via n8n + Google Sheets and deployed a public-facing frontend on GitHub Pages.

Applied AI Engineer · [Brane Enterprises](#)

Jan 2023 – Nov 2023 ·
Hyderabad, India (On Site)

- Deployed LLM-powered enterprise AI assistants for knowledge retrieval and document workflows across **4 business units**, reducing hallucination-related errors by **24%** through structured retrieval pipeline design.
- Designed and automated ETL pipelines processing **500K+ enterprise documents** of structured and unstructured data, reducing downstream inference inconsistency by an estimated **~30%**.
- Delivered API integrations and reporting dashboards adopted by **3 non-technical teams**, cutting average information-retrieval time from ~15 minutes to under 2 minutes per query.

EDUCATION

AI Product Management Specialization ·

May 2025 – Sept 2025

[Duke University](#)

University-level specialization covering AI strategy, product lifecycle, and responsible AI deployment.

M.S., Information Technology · GPA 4.0 / 4.0 ·

Jan 2024 – May 2025

[Florida State University](#)

Relevant coursework: NLP, Machine Learning Systems, Applied AI, Data Engineering

B.E., Electronics Engineering ·

Aug 2019 – May 2023

[Chaitanya Bharathi Institute of Technology](#)

Relevant coursework: MATLAB, Machine Learning in Power Systems, Python, Signals and Systems

Publication: "Power System Fault Detection Using Machine Learning" peer-reviewed journal, 2023

CERTIFICATIONS

Salesforce Certified AI Specialist · Salesforce
Salesforce Certified AI Associate · Salesforce
Model Context Protocol (MCP) Certification · Hugging Face

PROJECTS

- Mistsplitter - Governed AI Orchestration Platform for FintechFeb 2026 – Apr 2026
- Architected a policy-governed, MCP-native AI orchestration platform for fintech fraud, AML, and KYC - **7 specialized agents**, append-only audit trail, role-based tool permissions, and mandatory human review gates throughout.
 - Established **288 tests across 30 suites** covering agents, workflow runtime, API routes, MCP permissions, and audit integrity; TypeScript strict mode, zero *any* across 8 monorepo packages.
 - Exposed 18 typed MCP tools across 3 permission tiers; REST API + Next.js operations dashboard + CLI - built to plug directly into existing fintech infrastructure.

- Reliquary of Truth - Proof-Gated AI Software Engineering SystemNov 2025 – Jan 2026
- Reduced governance overhead by automating policy checks, security scanning, and proof bundling; cut manual review steps from 6 to 2 per delivery cycle across the audit pipeline.
 - Implemented policy/risk gates, vulnerability checks, and audit bundling for governance-aware, traceable AI-generated code workflows at enterprise scale.

- Neurobud - Production AI Mental Wellness PlatformSep 2025 – Nov 2025
- Shipped a **sub-3 second** production inference pipeline using Python, FastAPI, PostgreSQL, and vector-based retrieval for a healthcare-focused AI platform.
 - Implemented auth, rate-limiting, logging, and monitoring controls; modular backend architecture built for future analytics and reporting expansion.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
AI / ML	LLMs, RAG, Prompt Engineering, Fine-Tuning, NLP, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Classification, TF-IDF
Frameworks	FastAPI, Flask, Next.js, React, Node.js, scikit-learn, XGBoost, TensorFlow, PyTorch, Hugging Face Transformers
LLM / AI Tools	OpenAI, Azure OpenAI, LangChain, MCP (Model Context Protocol), n8n, Embeddings, pgvector, Pinecone, RAG Pipelines
Cloud / DevOps	Azure, Terraform (IaC), GitHub Actions, CI/CD, Docker, REST APIs, Git
Databases	PostgreSQL, MySQL, MongoDB, Prisma ORM, ETL Pipelines, Pandas, NumPy